

Adaptive Multi-Modal Fusion Transformer

for Indian Sign Language Recognition

Week 1 Literature Review

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Presentation Roadmap

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Indian Sign Language Recognition

Indian Sign Language is an important communication method for deaf and hard-of-hearing people.

- Communication barriers occur because many people do not know sign language.
- Artificial Intelligence, Computer Vision and Deep Learning can help build automated sign recognition systems.
- The aim is to create an intelligent system that understands visual, spatial and temporal information from videos.
- The system should recognize Indian Sign Language gestures accurately.

Video Input → Visual Features → Temporal Understanding → ISL Gesture Output

Problem Statement and Motivation

Problem

There is a communication gap between sign language users and people who do not understand sign language.

Major Challenges

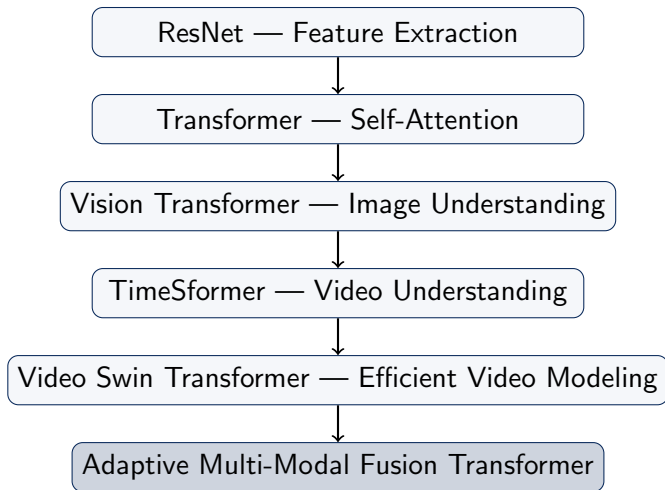
- Hand shape variation
- Gesture motion changes
- Facial expressions
- Background noise
- Similar-looking gestures

Objective

To build an AI-based system that can recognize Indian Sign Language gestures from videos.

Why Deep Learning?

Deep learning can learn visual patterns, motion information and complex relationships from sign language videos.



Why ResNet was introduced

Deep neural networks face degradation problems. When the number of layers increases, training becomes difficult and training error may increase.

Key Idea

- Residual learning
- Skip connections
- Smooth flow of information
- Better gradient flow

Relevance to ISL

ResNet can be used for feature extraction from sign language image frames.

Input → Conv Layers → Skip Connection →
Output

Transformer

Why Transformer was introduced

Traditional RNN and LSTM models struggle to capture long-range dependencies in sequences.

Core Concepts

- Self-attention
- Query, Key and Value
- Positional encoding
- Parallel processing

Relevance to ISL

A sign gesture is a sequence of frames. Transformers help understand relationships between different frames.

Vision Transformer (ViT)

Main Idea

Vision Transformer extends the Transformer architecture to image processing.

- The image is divided into uniformly sized patches.
- Each patch is converted into an embedding.
- Positional encoding is added to preserve location information.
- Transformer encoder layers process the patch sequence.
- A classification token represents the full image.

Relevance to ISL

ViT helps capture global relationships between hand regions, palm orientation and body posture.

Purpose

TimeSformer is a Transformer-based architecture designed for video understanding.

How it Works

- Spatial attention learns information inside each frame.
- Temporal attention learns information between frames.
- Divided attention separates space and time.

Relevance to ISL

It is useful for capturing hand movements and temporal dynamics of gesture sequences.

Video Swin Transformer

Main Idea

Video Swin Transformer uses shifted window attention instead of applying attention to all tokens.

- Reduces computational complexity.
- Uses hierarchical video representation.
- Learns local and global video features.
- Works well for action recognition, gesture recognition and sign language interpretation.

Key Advantage

More efficient and scalable than TimeSformer for larger video datasets.

Comparison Between CNNs and Transformers

CNNs

- Good at local visual feature learning.
- Useful for small datasets.
- Strong for image feature extraction.
- Weak in long-range dependency modeling.

Transformers

- Good at long-range relationship learning.
- Uses self-attention.
- Needs more data and computation.
- Suitable for spatial-temporal gesture understanding.

For ISL: Transformers are useful because gestures contain many interdependent frames.

Sign Language Recognition Research

Current Research Direction

Recent sign language recognition systems use Transformer-based architectures along with multimodal fusion techniques.

- Early methods used CNNs and RNNs for feature extraction and sequence modeling.
- Modern methods use Vision Transformer, TimeSformer, Video Swin Transformer and multimodal fusion.
- Sign language recognition uses hand landmarks, facial expressions, body posture and motion cues.

Remaining Challenges

Limited datasets, different signing styles, hand occlusion, background noise, motion blur and gesture similarity.

Why Self-Attention is Better Than Average Pooling

Average Pooling

- Treats all video frames equally.
- Gives the same weight to every frame.
- May reduce the importance of key gestures.

Self-Attention

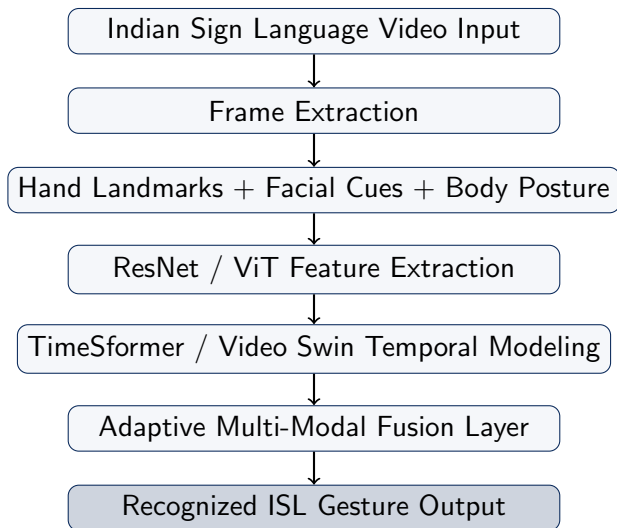
- Finds the most informative frames.
- Gives more importance to critical hand movements.
- Focuses on facial expressions and transitions.
- Improves video sequence understanding.

Self-attention provides a more robust solution for complex sign language gestures.

Comparative Summary

Model	Main Idea	Strength	Use in ISL
ResNet	Residual learning and skip connections	Strong image feature extraction	Frame-level hand feature extraction
Transformer	Self-attention mechanism	Captures long-range dependencies	Gesture sequence understanding
ViT	Image patches as tokens	Captures global image relationships	Hand shape and posture recognition
TimeSformer	Spatial and temporal attention	Learns motion over video frames	Gesture movement modeling
Video Swin Transformer	Shifted window attention	Efficient and scalable video learning	Large-scale ISL video recognition

Proposed Future Direction



Week 1 Learning Outcome

The literature review gave a basic understanding of deep learning and Transformer-based architectures for Indian Sign Language Recognition.

- ResNet supports effective feature extraction.
- Transformers model complex long-range relationships.
- Vision Transformer adapts Transformer concepts to image processing.
- TimeSformer and Video Swin Transformer extend these ideas to video understanding.
- These architectures are important for developing an Adaptive Multi-Modal Fusion Transformer.

Thank You